

What generates a CTMC ?

- ▶ $P(t + l) = P(t)P(l)$.
- ▶ In DTMC, we could use P to generate the chain on Matlab.
- ▶ What about CTMC ? Can we use $P(t)$?
- ▶ What is $\lim_{h \rightarrow 0} P(h)$?
- ▶ What is $\frac{dP(h)}{dh}$ evaluated at $h = 0$?

What generates a CTMC ?

- ▶ Lets look at $\frac{dP(h)}{dh}|_{h=0} = \lim_{h \rightarrow 0} \frac{P(h) - P(0)}{h} = \lim_{h \rightarrow 0} \frac{P(h) - I}{h}$.
- ▶ Define $Q := \lim_{h \rightarrow 0} \frac{P(h) - I}{h}$
- ▶ **Does it always exist ?** Yes! (Proposition 2.2 and 2.4 (Anderson))
- ▶ Q has terms of the form q_{ii} and q_{ij} for $i, j \in \{1, 2, \dots, M\}$.
- ▶ $q_{ii} = \frac{dp_{ii}(h)}{dh}|_{h=0}$. Similarly $q_{ij} = \frac{dp_{ij}(h)}{dh}|_{h=0}$

What generates a CTMC ?

Theorem

Let $P(t)$ be a transition function. Then the generator matrix $Q = \lim_{h \rightarrow 0} \frac{P(h) - I}{h}$ exists.

Theorem

$P(Y_t > u | X(t) = i) = e^{-a_i u}$ where $a_i > 0$.

Theorem

(Proposition 2.8 Anderson)

$P(Y_t > u | X(t) = i) := e^{q_{ii} u}$, i.e., $q_{ii} = -a_i$.

$P(X(t + Y_t) = j | X(t) = i) = \frac{q_{ij}}{|q_{ii}|}$.

Q generates the CTMC

- ▶ Cannot generate CTMC directly from $P(t)$.
- ▶ From $P(t)$, obtain Q using $Q = \frac{dP(h)}{dh} \big|_{h=0}$
- ▶ Consider Y_t when $X(t) = i$.
- ▶ Now use the following theorem for generating the CTMC on a computer

Theorem

(Proposition 2.8 Anderson: we won't see proof)

$P(Y_t > u | X(t) = i) := e^{q_{ii}u}$, i.e., $q_{ii} = -a_i$.

$P(X(t + Y_t) = j | X(t) = i) = \frac{q_{ij}}{|q_{ii}|}$.

Properties of Q

Theorem

$P(Y_i > u | X(t) = i) := e^{q_{ii}u}$, i.e., $q_{ii} = -a_i$.

$P(X(t + Y_t) = j | X(t) = i) = \frac{q_{ij}}{|q_{ii}|}$ where $q_{ij} \geq 0$.

- ▶ q_{ii} is negative. $q_{ii} = -\sum_{j \neq i} q_{ij}$.
- ▶ $|q_{ii}|$ is the exponential rate at which you leave state i .
- ▶ q_{ij} is the exponential rate at which you leave state i to go to state j .
- ▶ minimum of exponentials is exponential with aggregated rate.
- ▶ This justifies the rate of leaving state i to be $\sum_{j \neq i} q_{ij}$.

Equivalent definition of a CTMC using Q

- ▶ in the CTMC, you stay in state i for a random duration that has exponential($|q_{ii}|$) distribution.
- ▶ From i , you will move to state j with probability $\frac{q_{ij}}{|q_{ii}|}$.
- ▶ Equivalently, in state i , you have $M - 1$ exponential(q_{ij}) clocks for $j = 1, 2, \dots, i - 1, i + 1, \dots M$.
- ▶ You move to that state whose clock rings first!

Kolmogorov forward/backward equations CTMC

$$\blacktriangleright \frac{dP(t)}{dt} = \lim_{s \rightarrow 0} \frac{P(t+s) - P(t)}{s}$$

$$\blacktriangleright \frac{dP(t)}{dt} = P(t) \lim_{s \rightarrow 0} \frac{P(s) - I}{s}$$

$$\blacktriangleright \frac{dP(t)}{dt} = P(t)Q.$$

$\blacktriangleright P(t) = e^{tQ}$ satisfies the above. (Calculus of Matrix exponentials)

$$\blacktriangleright P(t) = e^{tQ} := I + tQ + \dots + \frac{(tQ)^n}{n!} \dots$$

Example: Poisson process $N(t)$ as a CTMC

- ▶ States $\mathcal{S} = \mathbb{Z}_{\geq 0}$.
- ▶ Why is it a Markov process / Markov property satisfied?
- ▶ $P(N(t) = k | N(t_1) = k_1, \dots, N(t_m) = k_m) = P(N(t) = k | N(t_m) = k_m)$?
- ▶ $P(N(t) = k | N(t_1) = k_1, \dots, N(t_m) = k_m) = P(N(t - t_k) = k - k_m)$. Therefore the above is true.
- ▶ $p_{ij}(t) = P(N(t) = j | N(0) = i)$. $\max(j - i, 0)$ arrivals in time t .
- ▶ We know that this has Poisson distribution.
- ▶ How does $P(t)$ look for a Poisson process ?

Example: Poisson process $N(t)$ as a CTMC

- ▶ How does $P(t) = [[P(N(t) = j | N(0) = i)]]$ look ?
- ▶ Entries below the diagonal are zero.
- ▶ Diagonal entries have the value $e^{-\lambda t}$
- ▶ ij th entry above the diagonal has the value $e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!}$

Example: Poisson process $N(t)$ as a CTMC

- ▶ How does $Q = \frac{dP(h)}{dh}|_{h=0}$ look ?
- ▶ ij th entry above the diagonal $p_{ij}(t) = e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!}$
- ▶ what is $\frac{d}{dt} \left(e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!} \right) |_{t=0}$?
- ▶ If $j - i = 1$, then $\frac{d}{dt} \left(e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!} \right) = \lambda$.
- ▶ If $j - i > 1$, then $\frac{d}{dt} \left(e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!} \right) = 0$.
- ▶ How does Q for Poisson process look like ?
- ▶ $P(t) = e^{tQ} = I + tQ + \dots + \frac{(tQ)^n}{n!} + \dots$

Example 3: Binomial process as a DTMC

► DO IT YOURSELF!

Limiting probabilities

$$\blacktriangleright P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0.6 & 0.4 \\ 0.6 & 0.4 & 0 \end{bmatrix} \quad P^5 = \begin{bmatrix} .06 & .3 & .64 \\ .18 & .38 & .44 \\ .38 & .44 & .18 \end{bmatrix} \quad P^{30} = \begin{bmatrix} .23 & .385 & .385 \\ .23 & .385 & .385 \\ .23 & .385 & .385 \end{bmatrix}$$

$$\blacktriangleright P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \quad \lim_{n \rightarrow \infty} P^n = \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$$

$\blacktriangleright$ What is the interpretation of $\lim_{n \rightarrow \infty} p_{ij}^{(n)} = [\lim_{n \rightarrow \infty} P^n]_{ij}$?

$\blacktriangleright$ $\alpha_j = \lim_{n \rightarrow \infty} p_{ij}^{(n)}$ denotes the probability of being in state j after a large time from starting in state i .

$\blacktriangleright$ For an M state DTMC, $\alpha = (\alpha_1, \dots, \alpha_M)$ denotes the limiting distribution.

$\blacktriangleright$ How do we obtain the limiting distribution α ? Does it always exist?

Stationary distribution

The **stationary distribution** of a Markov chain is defined as a solution to the equation $\pi = \pi P$.

- ▶ πP is essentially the p.m.f of X_1 having picked X_0 according to π .
- ▶ $\pi = \pi P$ says that, if the initial distribution is π , then the distribution of X_1 is also π .
- ▶ Continuing this argument, the p.m.f of X_n for any n is π and there is no dependence on the starting state.
- ▶ MCMC algorithms use this idea (at stationarity successive states of the Markov chain have p.m.f π) to sample from target distribution π .

Limiting vs Stationary distribution

- ▶ Obtain stationary distribution for the Markov Chain with transition probability $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0.6 & 0.4 \\ 0.6 & 0.4 & 0 \end{bmatrix}$
- ▶ The limiting distribution α need not exist for some Markov chains, but the stationary distribution π exists. For example for $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
- ▶ The limiting distribution if it exists, is same as the stationary distribution, i.e. $\alpha_i = \pi_i$ for all i .
- ▶ For a CTMC, we know that $\frac{dP(t)}{dt} = P(t)Q$. When $\lim_{t \rightarrow \infty} P(t) = \Pi$, this means that at stationarity $\frac{dP(t)}{dt} = 0$. Therefore we have $\pi Q = 0$ in case of CTMC.